Ch7 - Optimization - Mathematical Background

An optimization problem can be stated as:

Find $\underline{x}$, which minimizes or maximizes $f(\underline{x})$:

subject to:

$$d_i(\underline{x}) \leq a_i \quad i = 1, 2, \cdots m \rightarrow \text{inequality constraints}$$

$$e_i(\underline{x}) = b_i \quad i = 1, 2, \cdots p \rightarrow \text{equality constraints}$$

$\underline{x}$: n dimensional vector

$f(\underline{x})$: objective function

a_i, b_i : constants

- If $f(\underline{x})$ and constraints are linear $\Rightarrow$ linear programming
- If " is quadratic and constraints are linear $\Rightarrow$ quadratic programming

- Constrained optimization:

  Degrees of freedom $= n - p - m$

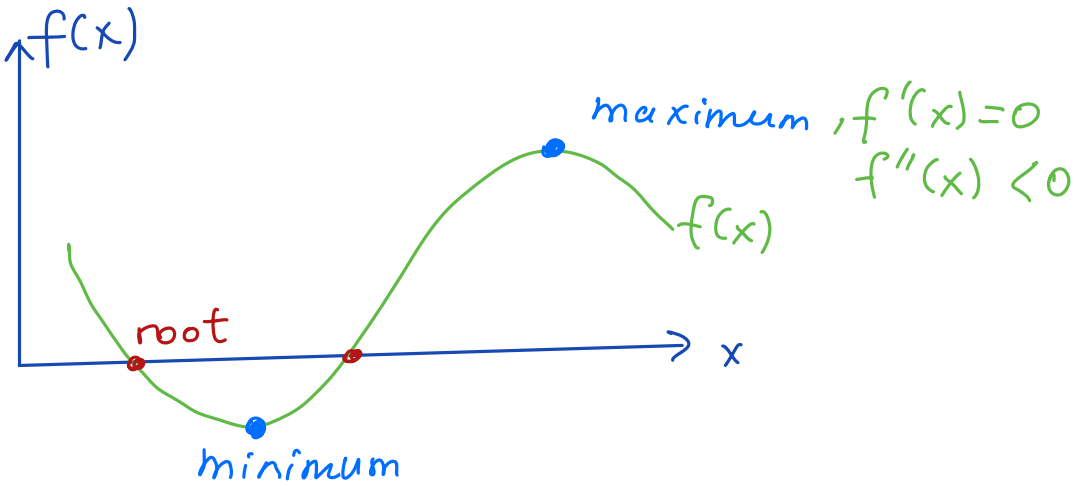
  To obtain a solution $p + m \leq n$

  If $p + m > n \Rightarrow$ the problem is overconstrained

- Unconstrained optimization
- 1D-problems: $f(x)$
- Multi-dimensional problems: $f(\underline{x})$, $\underline{x} \in \mathbb{R}^n$

* In the general sense, optimization is the process of creating something that is as effective as possible.
* Engineers always confront optimization problems that attempt to balance performance and limitations.

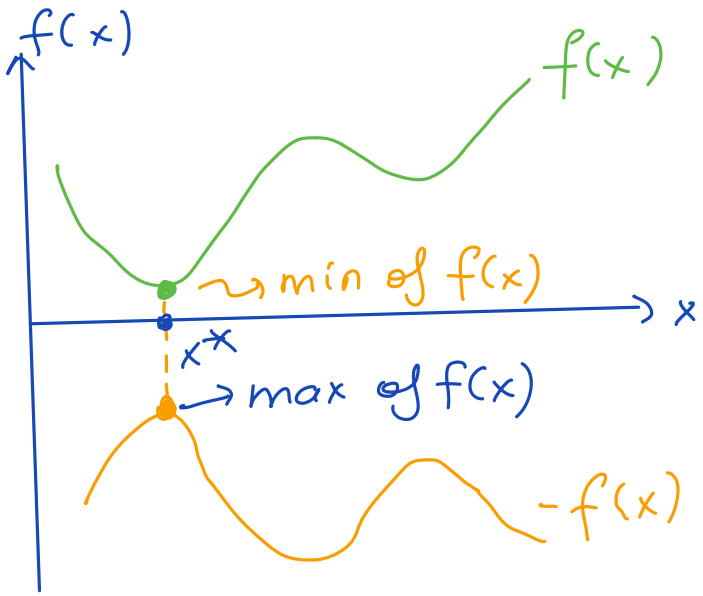
mathematically: find the minima or maxima of a function.



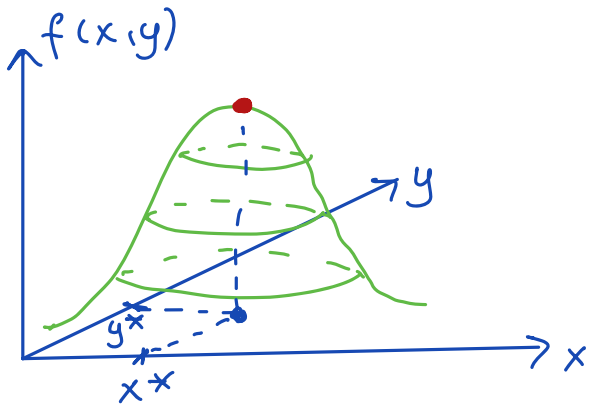
Relationship btwn roots and optima:

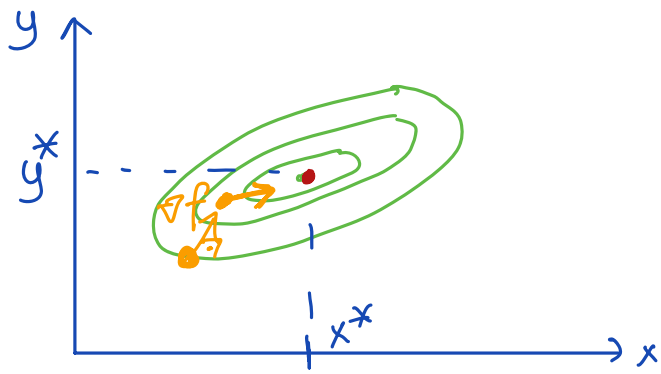
$f'(x)=0$ (roots of $f'(x)$)

1D optimization



2D optimization





Finding the min or max are identical

x^* minimizes $f(x)$

x^* maximizes $-f(x)$

$f(x) \rightarrow \frac{df}{dx}$

$f(x,y) \rightarrow \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}_{2\times 1} = \nabla f$ (Gradient of f)

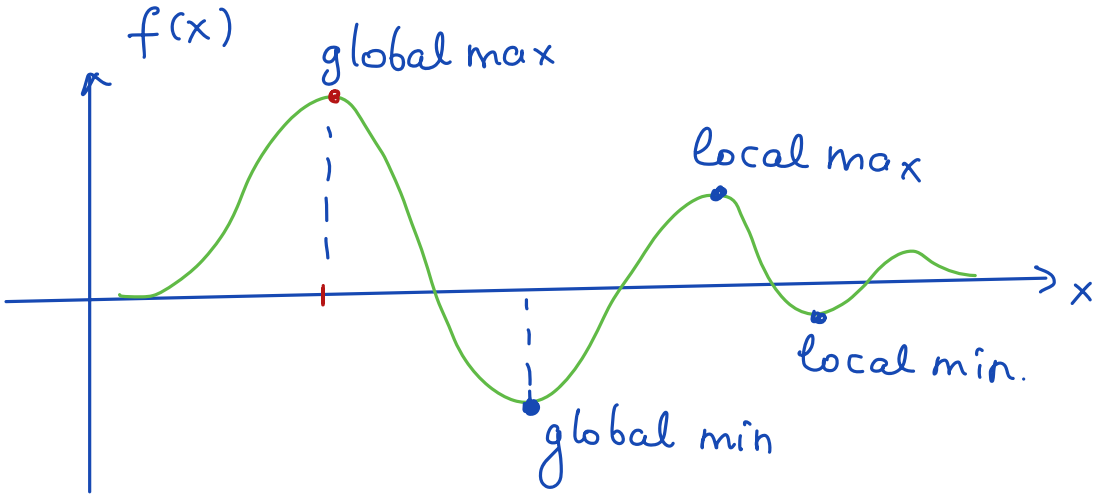
∇f shows the direction of maximum increase on the x-y plane.

One-Dimensional Optimization

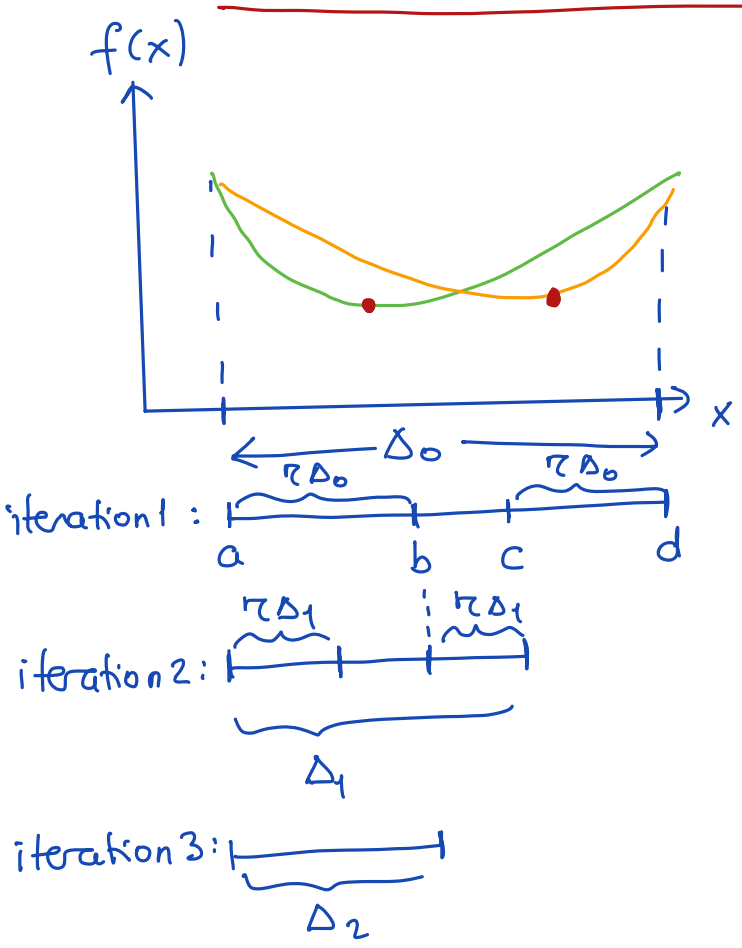
Global Optimum : the very best solution.

Local Optimum : though not the best, better than the neighbors.

* We are interested in finding global optimum



Golden Section Method (Michael Zibulevsky) - YouTube



- Evaluating a function at a single intermediate value will not whether the min. is in the left half or right half.
- Evaluate at 2 intermediate pts. and get an idea of the slope

if $f(b) < f(c)$
choose ac
else
choose bd

- we want to carry one intermediate pt. to the next iteration.

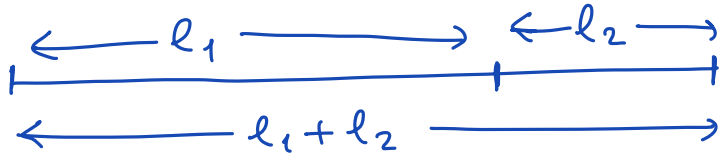
$$\Delta_1 = \Delta_0 - \tau\Delta_0 = (1-\tau)\Delta_0$$

$$\tau\Delta_0 = \Delta_1 - \tau\Delta_1 = (1-\tau)\Delta_1 = (1-\tau)^2\Delta_0$$

$$\tau = (1-\tau)^2 \Rightarrow \tau = \frac{3-\sqrt{5}}{2} = 0.3820$$

Most efficient method to find the min. since we have only function evaluation (no derivative evaluation)

Golden-Ratio :



Euclid's golden ratio : divide a line

$$\frac{l_1+l_2}{l_1} = \frac{l_1}{l_2}$$

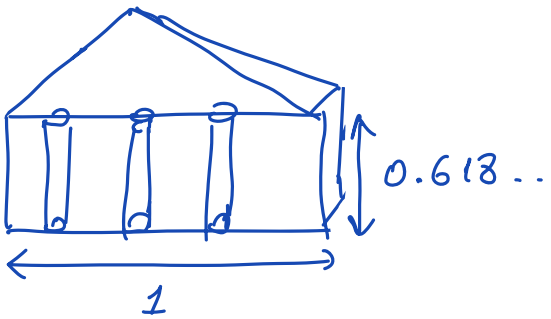
Multiply by $\frac{l_1}{l_2}$

$$\frac{l_1}{l_2} + 1 = \frac{l_1^2}{l_2^2} \quad , \text{let } \frac{l_1}{l_2} = \phi$$

$$\phi^2 - \phi - 1 = 0$$

$$\Rightarrow \phi = \frac{1+\sqrt{5}}{2} = 1.618$$

$\phi - 1 = 0.618\ldots$ is the Golden ratio.



- Partheron in Athens uses the Golden Ratio
- Golden ratio has been found to be aesthetically pleasing in Western cultures.

Golden-Section Search:

Start by defining an interval that contains a single answer- (interval contains a single min.)

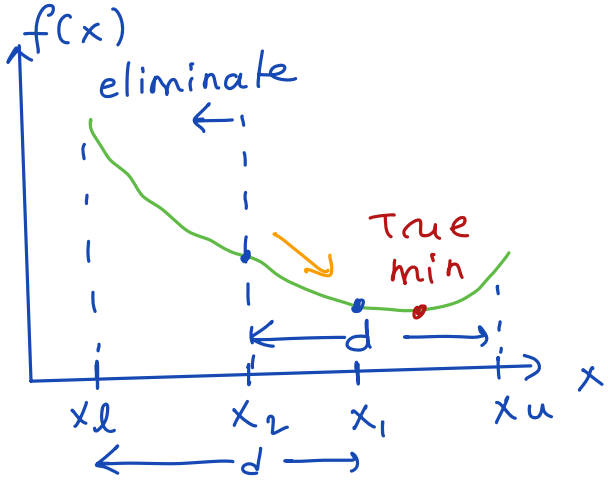
x_l : lower bound

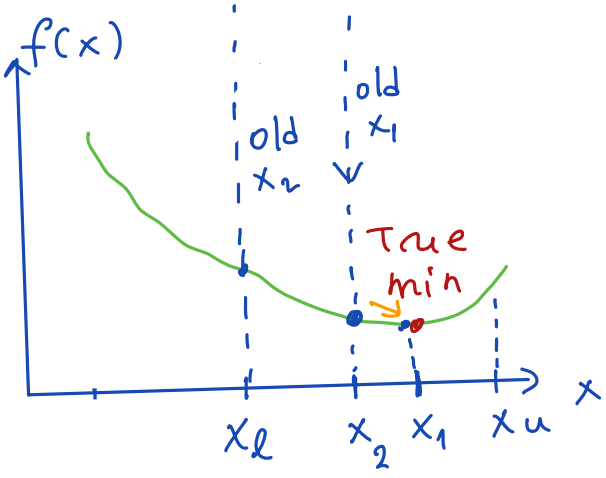
x_u : upper bound

initial step : intermediate pts : x_1 and x_2

$$\left.\begin{aligned} x_1 &= x_l + d \\ x_2 &= x_u - d \end{aligned}\right\} \quad d = \underbrace{(\phi - 1)}_{\text{golden ratio}}(x_u - x_l)$$

(Goal : determine two intermediate pts chosen by the Golden ratio)





initial step : choose two interior pts (x_1 and x_2) according to the golden ratio.

second step : define a new interval that encompasses the optimum.

Second Step : Evaluate the f^n at 2 intermediate pts.

1) If $f(x_1) < f(x_2)$ then $f(x_1)$ is the minimum.

- eliminate the region from x_l to x_2 since it doesn't contain the min. (left of x_2)
- replace x_l by x_2
- old x_1 becomes x_2
- $x_1 = x_l + d_{new}$ $\quad (d_{new} = (\phi - 1)(x_u - x_l))$

2) if $f(x_1) > f(x_2)$ then $f(x_2)$ is the min.

- Eliminate the right of x_1 to x_u
- Replace x_u with x_1
- old x_2 becomes x_1
- $x_2 = x_u - d_{new}$ $\quad (d_{new} = (\phi - 1)(x_u - x_l))$

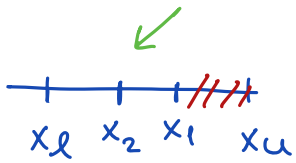
- At each iteration, the interval is reduced by a factor of about 61.8%
- After 20 iterations interval is shrunk to about 0.0066% of its initial length.

EX Find the min. of $f(x) = \frac{x^2}{10} - 2\sin x$, from $x_l = 0$ to $x_u = 4$

$d = 0.618(x_u - x_l) = 0.618(4) = 2.4721$

$x_1 = x_l + d = 2.4721$, $f(x_1) = -0.63$

$x_2 = x_u - d = 1.5279$, $f(x_2) = -1.7647$



Since $f(x_2) < f(x_1) \Rightarrow$ best estimate of min. is $f(x_2)$, at $x_2 = 1.5279$

iter. 1

$\Rightarrow$ min. is in x_l, x_2, x_1

$\Rightarrow x_l^{new} = 0$, $x_u^{new} = x_1 = 2.4721$

$\Rightarrow x_1^{new} = x_2 = 1.5279$, $f(x_1^{new}) = -1.7647$

$\rightarrow$ update $d^{new} = 0.618(x_u^{new} - x_l^{new})$

$= 0.618(2.4721 - 0) = 1.5279$

$\rightarrow x_2^{new} = x_u^{new} - d^{new} = 2.4721 - 1.5279 = 0.9443$

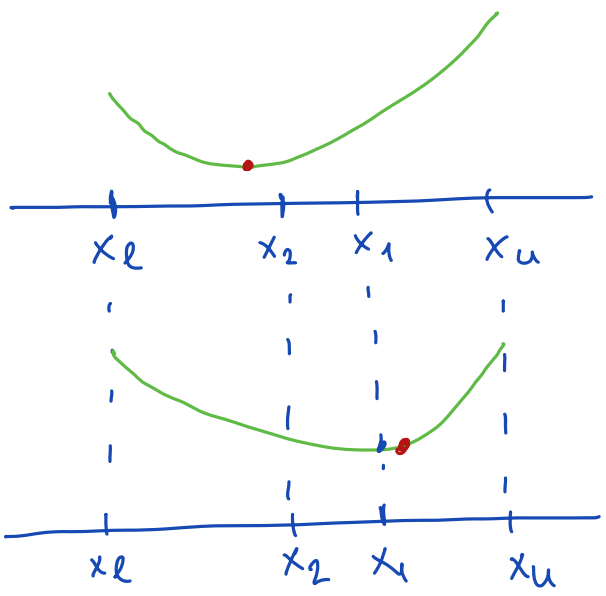
$\rightarrow f(x_2^{new}) = -1.5310$

$f(x_2^{new}) > f(x_1^{new}) \Rightarrow$ best min is at $x_1^{new} = 1.52$

(in interval x_2, x_1, x_u)

Upper Bound For the Error of Golden-Section Search

complete an iteration:



Case 1

- current best optimum at x_2
- real optimum is in lower interval (x_l, x_2, x_1)

Case 2

- current best optimum at x_1
- real optimum is in upper interval (x_2, x_1, x_u)

Case 1 & Case 2 are symmetric. Focus on case 2: (x_2, x_1, x_u)

$\Rightarrow$ if the true min. is at x_2 (far left) max. distance:

$$\Delta x_a = x_1 - x_2 \quad (x_1 \text{: current best})$$

$$= \underbrace{x_l + (\phi-1)(x_u - x_l)}_{x_1} - \underbrace{\left[x_u - (\phi-1)(x_u - x_l)\right]}_{x_2}$$

$$= (x_l - x_u) + 2(\phi-1)(x_u - x_l)$$

$$= (2\phi - 3)(x_u - x_l) = 0.2361\,(x_u - x_l)$$

$\Rightarrow$ if the true value of min is at x_u (far right)

$$\Delta x_b = x_u - x_1$$

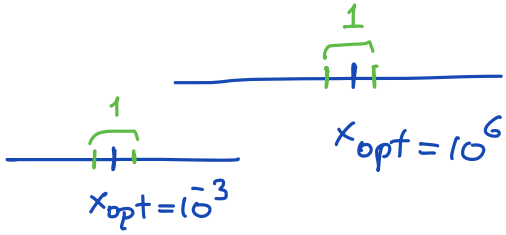
$$= x_u - \left[x_l + (\phi-1)(x_u - x_l)\right]$$

$$= (2-\phi)(x_u - x_l) = 0.3820\,(x_u - x_l)$$

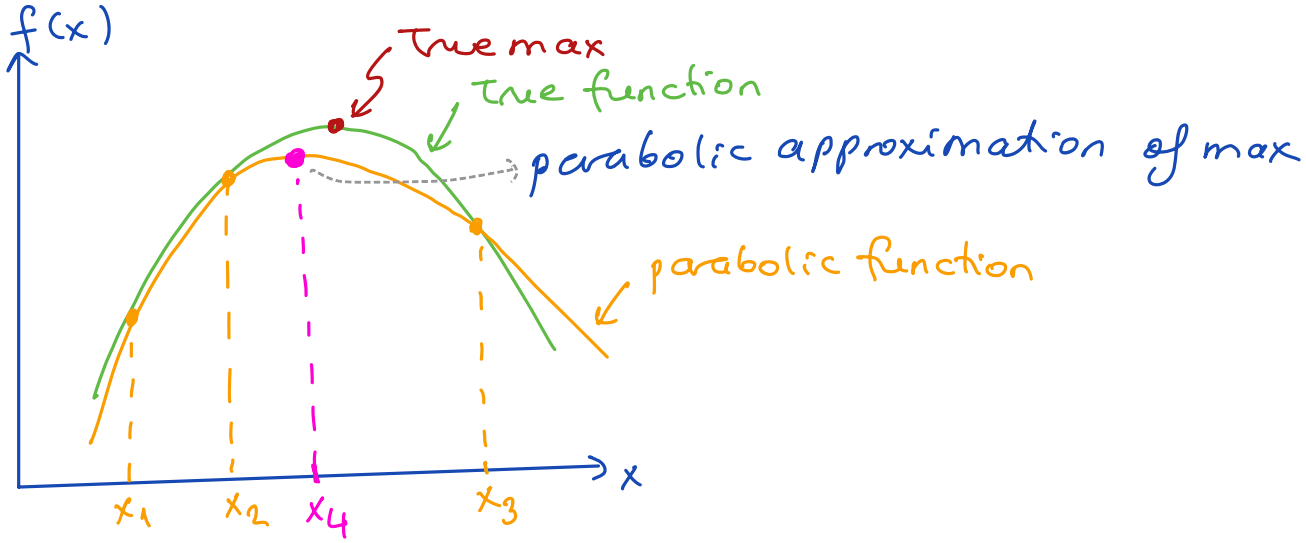
Normalize to get a relative error at each iteration:

$$\varepsilon_a = (2-\phi)\left|\frac{x_u - x_l}{x_{opt}}\right| \times 100\ \%$$

can be used as a terminating condition.



Parabolic Interpolation



- Given $(x_1, f(x_1))$, $(x_2, f(x_2))$ and $(x_3, f(x_3))$ fit a parabola to the 3 pts. (initial guesses)
- Differentiate the parabola, equate to zero, find x_4.

$$x_4 = x_2 - \frac{1}{2}\frac{(x_2-x_1)^2[f(x_2)-f(x_3)] - (x_2-x_3)^2[f(x_2)-f(x_1)]}{(x_2-x_1)[f(x_2)-f(x_3)] - (x_2-x_3)[f(x_2)-f(x_1)]} \quad (1)$$

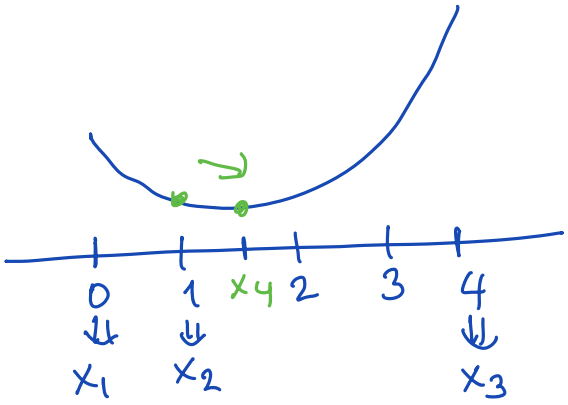
x_4: updated optimum of parabolic fit.

EX Approximate the min. of $f(x) = \frac{x^2}{10} - 2\sin x$

initial guesses $x_1 = 0$, $x_2 = 1$, $x_3 = 4$

$f(x_1) = 0$ $f(x_2) = -1.58$, $f(x_3) = 3.11$

from (1) $\Rightarrow$ $x_4 = 1.5$ $\rightarrow f(x_4) = -1.7691$



$f(x_4) < f(x_2)$, discard x_1

Set $x_1 = 1$, $x_2 = 1.5$, $x_3 = 4$ } in the next iteration

Newton's Method

(Open method to find the roots: $f(x)=0$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

)

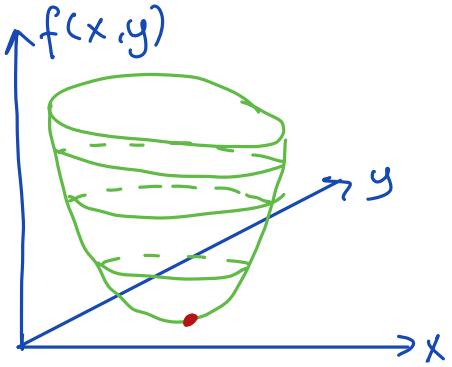
To find the optimum of $f(x)$

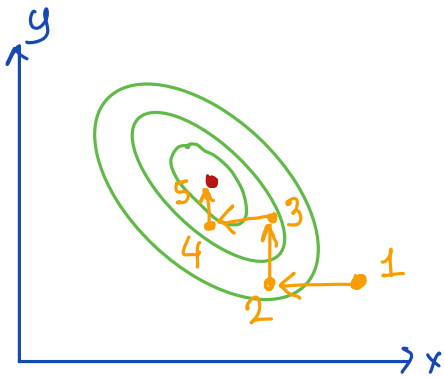
$$f'(x) = g(x) = 0$$

$$x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}$$

Multidimensional Optimization

Univariate Search Method



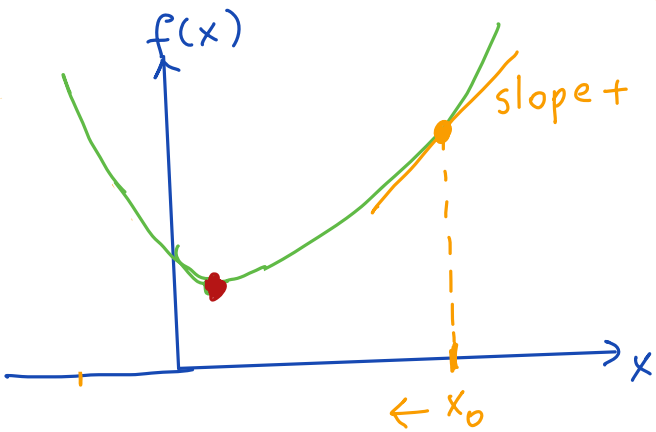


- Change one variable at a time keep the other variable constant.
- Sequence of 1D searches.

→ Start at pt 1, move along x-axis with y constant
→ next move along y axis with x constant
→ Repeat the above 2 steps.

Gradient Based Method

- use derivative



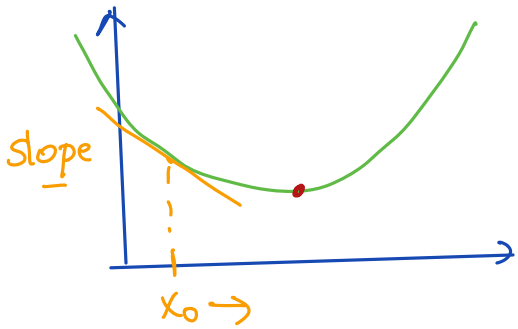
$f'(x_0) > 0$

- if slope > 0 ⇒ increasing x will increase f(x)
- if slope < 0 ⇒ decrease x, f(x) ↑
- if slope $= 0$, we're at optimum
  - → if $f''(x) > 0$ ⇒ local min
  - → if $f''(x) < 0$ ⇒ local max

gradient descent

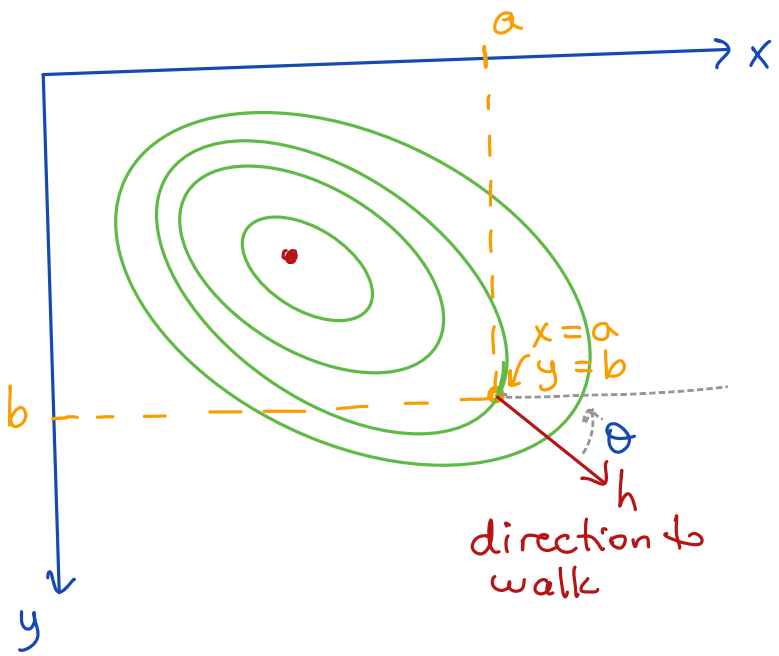
$$x_{i+1} = x_i - \gamma \left.\frac{df(x)}{dx}\right|_{x=x_i}$$

(γ: step size)



The Gradient

2D function $f(x,y)$



- climbing a mountain initial position (a,b)

The elevation along the direction of red arrow.

Directional Derivative

$$g'(0) = \frac{\partial f}{\partial x}\cos\theta + \frac{\partial f}{\partial y}\sin\theta$$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \cdot \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

Goal: To gain the most elevation with the next step. what's the direction steepest ascent?

Gradient: $\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j}$ (at $x=a$, $y=b$)

"del f"

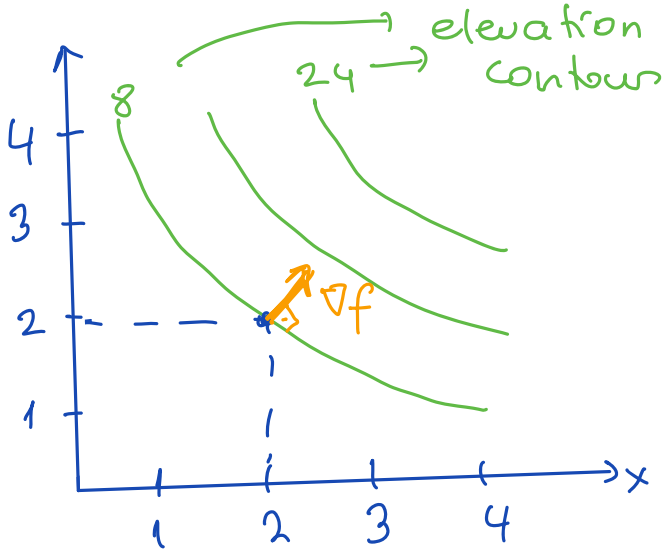
generalize $\nabla f(\underline{x}) = \begin{bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix}$

$\underline{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

* Gradient shows what direction to go **locally** and how much elevation we will gain by walking in that direction.

Ex Employ the gradient to evaluate the steepest ascent direction for:

$f(x,y) = xy^2$ at the point $(2,2)$



Gradient is always perpendicular to elevation contour.

$f(2,2) = 2(2)^2 = 8$

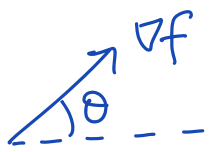
$$\frac{\partial f}{\partial x} = y^2 \Big|_{y=2} = 4$$

$$\frac{\partial f}{\partial y} = 2xy \Big|_{\substack{x=2 \\ y=2}} = 8$$

$$\text{Gradient} = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$

$\nabla f = 4\,\underline{i} + 8\,\underline{j}$

Direction:



$\theta = \arctan\left(\frac{8}{4}\right) = 63^\circ$ (direction)

$\|\nabla f\| = \sqrt{4^2+8^2} = 8.94$

we will gain 8.94 units of elevation rise if you walk 1 unit along the gradient. (steepest path)

* As you walk, direction and magnitude of gradient will change. You have to modify your climbing at each step.

The Hessian

1D problems: both 1st and 2nd orders are useful for searching the optima.

1st derivative: a) the direction steepest ascent

b) tells if you have reached the optimum

2nd der.: $f''(x) < 0 \Rightarrow$ max pt.

$f''(x) > 0 \Rightarrow$ min pt.

2D functions: $f(x,y)$

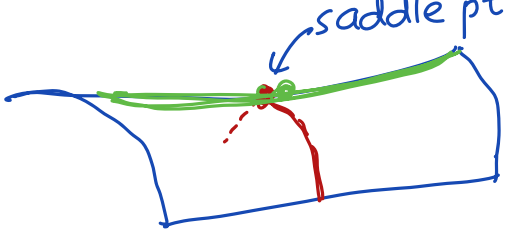
Hessian of f:

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

$$|H| = \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

3 cases:

- If $|H| > 0$ and $\frac{\partial^2 f}{\partial x^2} > 0 \Rightarrow f(x,y)$ has a local min
- If $|H| > 0$ and $\frac{\partial^2 f}{\partial x^2} < 0 \Rightarrow$ " " max
- If $|H| < 0$, then $f(x,y)$ has a saddle point (neither max nor min)



$$\frac{\partial f}{\partial x} = \frac{f(x+\delta x, y) - f(x-\delta x, y)}{2\delta x}$$

δ : small perturbation

$$\frac{\partial f}{\partial y} = \frac{f(x, y+\delta y) - f(x, y-\delta y)}{2\delta y}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{f(x+\delta x, y) - 2f(x,y) + f(x-\delta x, y)}{\delta x^2}$$

$$\frac{\partial f^2}{\partial y^2} = \frac{f(x, y+\delta y) - 2f(x,y) + f(x, y-\delta y)}{\delta y^2}$$

$$\frac{\partial^2 f}{\partial x\, \partial y} = \frac{f(x+\delta x, y+\delta y) - f(x+\delta x, y-\delta y) - f(x-\delta x, y+\delta y) - f(x-\delta x, y-\delta y)}{4\delta x \delta y}$$

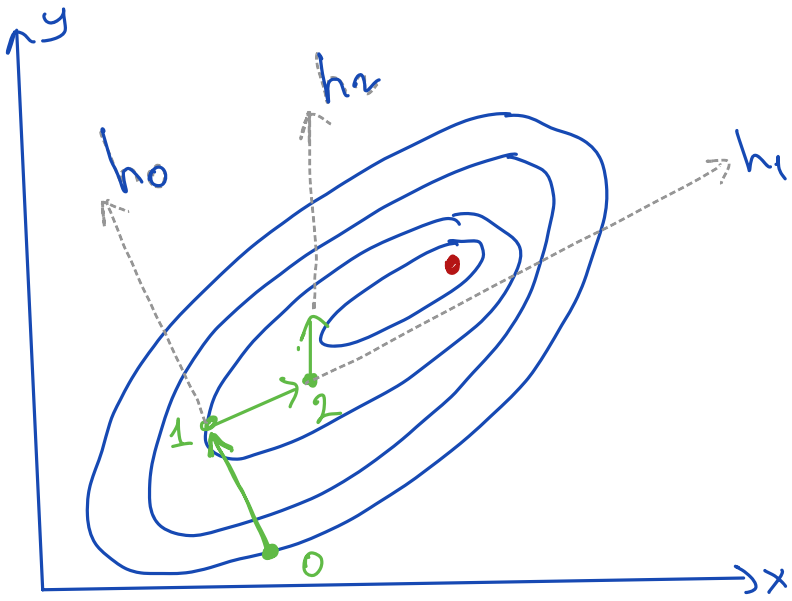
- If possible use exact (analytical) derivatives.

Steepest Ascent Method

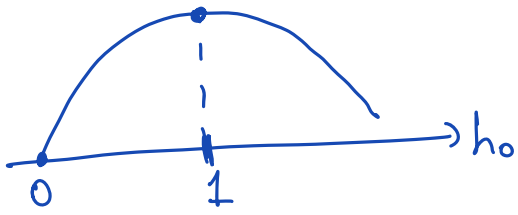
- Determine the maximum slope at current position.
- Walk a short distance in that direction.
- Stop, reevaluate the gradient and walk a short distance
- Repeat the process, until you reach maximum.

→ Above approach is not practical because of continuous reevaluation of the gradient.

Preferred version: move in a fixed path along the initial gradient direction until $f(x,y)$ stops increasing. Then repeat the process.



Start at point 0, move in the direction of gradient (h_0) until a max. is reached along h_0.



(1) Determine the "best" direction

(2) " " "best" value in that direction.

How to transform a function $f(x,y)$ into a function of h along the gradient direction:

Start at (x_0, y_0) and move in the direction of h:

$$x = x_0 + \left.\frac{\partial f}{\partial x}\right|_{x=x_0,\, y=y_0} h$$

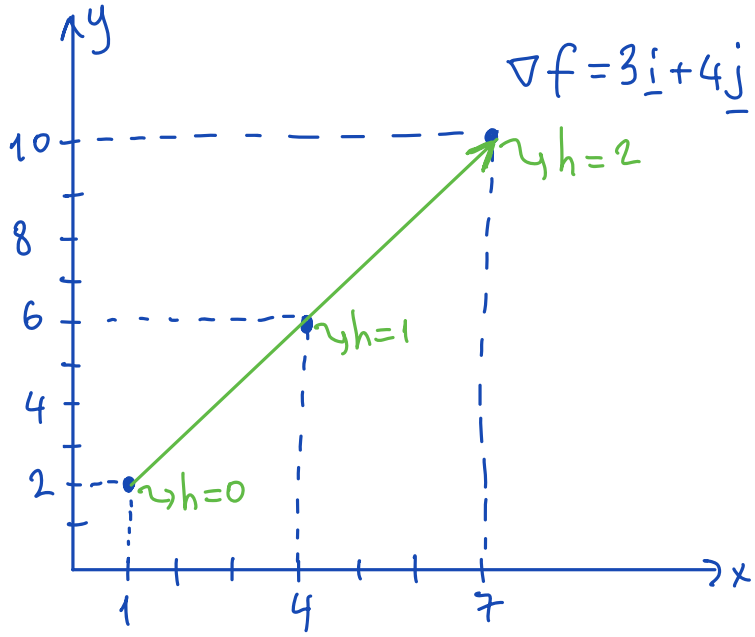
$$y = y_0 + \left.\frac{\partial f}{\partial y}\right|_{\substack{y=y_0 \\ x=x_0}} h$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + \left.\begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}\right|_{\substack{x=x_0 \\ y=y_0}} h$$

h: distance along h axis.

Let $x_0 = 1$, $y_0 = 2$, $\nabla f = 3\underline{i} + 4\underline{j}$

$$\left.\begin{aligned} x &= 1 + 3h \\ y &= 2 + 4h \end{aligned}\right\}$$ coordinate of any pt. along h axis


y
x
10
8
6
4
2
1
4
7
$\nabla f = 3\underline{i} + 4\underline{j}$
$\nabla h = 2$
$\nabla h = 1$
$\nabla h = 0$